VELOOP REWARDS - User & Account Analytics Specification

Data Analyst Internship, Task 01 | Work Package B

Authenticity notice: Every identifier and field below is **Proposed** or **To Be Verified**. Nothing is claimed to exist in VELOOP today, and no simulated or real VELOOP statistics appear here.

1. Purpose

This specification defines the structure needed to follow a user from registration through login, onboarding and account changes. It covers four identifiers (B1), a field-level specification (B2), a lifecycle model and a list of data that should deliberately not be collected. It uses IDs and analytical attributes only, as the brief requires.

2. Key design decisions

- **user_id is the anchor.** Every other dataset joins to it. If it is not stable, retention, cohort and cross-feature analysis are unreliable (Gap G01).
- **transaction_id must be unique and idempotent.** Otherwise duplicate wallet credits cannot be detected (Gap G02).
- **Session ID is optional.** DAU, MAU and most KPIs work from events; session_id only adds session-level analysis, so it is Proposed rather than required.
- **Active means qualifying activity, not just login.** Which events qualify, and the length of the activity window, are open decisions for stakeholders. No target number is invented here.
- **Derived fields are computed in the analytics layer**, not asked of engineering (5 of 29 fields).
- **Privacy first.** Names, emails, phone numbers, exact location, raw IP and secrets are excluded (Section 6).

3. Identifier requirements (B1)

Identifier	Proposed Field	Definition	Why It Is Needed	Proposed Format	Rules	Generated By	Status
Stable user identifier	user_id	One permanent, anonymous key per VELOOP account	Joins users to every task, reward, wallet, withdrawal and referral record; needed for cohorts and retention	Opaque string (UUID or equivalent). Never an email, phone or name	Unique, never reused, never changed, never derived from personal data	Backend at registration	To Be Verified
Event identifier	event_id	One unique key per tracked event occurrence	Detects duplicate events, supports retries and lets engineering trace one event end to end	UUID generated when the event is created	Unique per event; a retry of the same event must reuse the same event_id	Client or backend that emits the event	Proposed
Session identifier	session_id	Key grouping events from one continuous app visit	Enables session length, events per session and session-level funnels	Opaque string; new value at each session start	Where appropriate only; DAU/MAU do not depend on it; session timeout rule to be agreed	Client or backend at login/app open	Proposed
Transaction identifier	transaction_id	One unique key per wallet transaction	Prevents double counting of credits, links reversals to originals, reconciles wallet vs rewards vs withdrawals	Opaque string generated server-side	Unique; idempotent write (same request cannot create two rows); reversals carry related_transaction_id	Backend wallet service	To Be Verified

4. User analytics field specification (B2)

29 fields across Registration, Login, Onboarding and Account Lifecycle. 'Conditional' means required only when the stated condition applies; 'Derived' means calculated in the analytics layer.

Group	Field	Type	Required?	Purpose	Source	Validation	Status
Registration	user_id	string	Yes	Stable key for the user	Backend	Non-null; unique; immutable	To Be Verified
Registration	registered_at	timestamp (UTC)	Yes	Registration date for cohorts and growth	Backend	ISO 8601 UTC; not in the future; not before product launch date	To Be Verified
Registration	registration_source	string (enum)	Yes	Acquisition channel analysis	Frontend / Backend	Must be in approved list; use 'unknown' rather than null	Proposed
Registration	registration_method	string (enum)	Optional	Compare sign-up routes without storing identity details	Backend	Approved list only; the identity value itself is never stored in analytics	Proposed
Registration	platform	string (enum)	Yes	Split behaviour by android / ios / web	Frontend	One of: android, ios, web (final list to be agreed)	Proposed
Registration	app_version	string	Yes	Find version-specific problems	Frontend	Matches version pattern, e.g. 1.4.2	Proposed
Registration	country_code	string	Optional	Coarse geographic distribution (dashboard 2)	Backend (derived)	ISO 3166-1 alpha-2; country level only, no precise location	Proposed
Registration	language_code	string	Optional	Content and support planning	Frontend	ISO 639-1	Proposed
Registration	referral_code_used	string	Optional	Links new user to a referral record	Backend	Must match an existing referral code, or null	To Be Verified
Registration	campaign_id	string	Optional	Marketing campaign attribution	Frontend / Backend	Approved campaign list or null	Proposed
Login	login_at	timestamp (UTC)	Yes	Return behaviour and last-login tracking	Backend	ISO 8601 UTC; not before registered_at	Proposed
Login	session_id	string	Optional	Session-level analysis where appropriate	Client / Backend	Unique per session; non-null if session tracking adopted	Proposed
Login	login_platform	string (enum)	Yes	Platform used at login (may differ from registration)	Frontend	Same list as platform	Proposed
Login	logout_at	timestamp (UTC)	Optional	Session length where logout exists	Frontend	Not before login_at; may be missing for timeouts	Proposed
Login	last_login_at	timestamp (UTC)	Derived	Latest login per user for lifecycle checks	Analytics layer	Equals max(login_at) per user	Proposed
Onboarding	onboarding_status	string (enum)	Yes	Funnel position of each user	Backend	not_started, in_progress, completed, skipped (final list to be agreed)	Proposed
Onboarding	onboarding_step	string	Optional	Find the exact step where users stop	Frontend	Must be in step list of current onboarding flow	Proposed
Onboarding	onboarding_started_at	timestamp (UTC)	Conditional	Time to start onboarding	Frontend	Required when status is not not_started; not before registered_at	Proposed
Onboarding	onboarding_completed_at	timestamp (UTC)	Conditional	Onboarding completion rate and duration	Backend	Required when status = completed; not before started_at	Proposed
Onboarding	first_activity_at	timestamp (UTC)	Derived	Time from registration to first qualifying activity	Analytics layer	Min timestamp of qualifying events; not before registered_at	Proposed
Onboarding	first_reward_credited_at	timestamp (UTC)	Derived	Milestone: first credited reward (not first activity)	Analytics layer	From first reward_credited event for the user	Proposed

Group	Field	Type	Required?	Purpose	Source	Validation	Status
Account Lifecycle	account_status	string (enum)	Yes	Current state of the account	Backend	Must be one of the approved statuses (see lifecycle sheet)	To Be Verified
Account Lifecycle	previous_status	string (enum)	Conditional	Needed to read status changes	Backend	Required on a status change; differs from account_status	Proposed
Account Lifecycle	status_changed_at	timestamp (UTC)	Conditional	When a status changed	Backend	Required on a status change; not before registered_at	Proposed
Account Lifecycle	status_change_actor	string (enum)	Conditional	Who caused the change: user, system or operations	Backend	user, system, operations	Proposed
Account Lifecycle	status_reason_category	string (enum)	Optional	Why the status changed, at category level only	Backend	Categories to be agreed with Operations; no free text with personal data	Proposed
Account Lifecycle	last_activity_at	timestamp (UTC)	Derived	Recency; basis for inactive-user KPI	Analytics layer	Max timestamp of qualifying activity events	Proposed
Account Lifecycle	account_closed_at	timestamp (UTC)	Conditional	Churn and closure timing	Backend	Required when status = closed	Proposed
Account Lifecycle	lifecycle_stage	string (derived)	Derived	Single label per user for dashboards	Analytics layer	Computed from the lifecycle rules sheet	Proposed

5. Account lifecycle model

States are proposed labels for analysis. Real status names must be mapped from VELOOP once confirmed by engineering (Assumption Register A003, A012).

State	Meaning	Entry Condition	Stored or Derived	Open Decision	Status
Registered	Account exists, onboarding not begun	user_registered received	Derived	-	Proposed
Onboarding	User is in the onboarding flow	onboarding_started and not onboarding_completed	Derived	Does skipping count as completed?	Proposed
Activated	User has done the activating action	Definition to be agreed (candidates: onboarding completed, first task completed, first reward credited)	Derived	Choose one definition; referral activation (A010) should use the same one	To Be Verified
Active	User performed a qualifying activity inside the activity window	At least one qualifying event in the window	Derived	Which events qualify; window length	Proposed
Inactive	No qualifying activity inside the window	No qualifying event in the window	Derived	Window length to be agreed; no target is invented here	Proposed
Restricted	Account limited by system or operations	account_status = restricted	Stored	Real status names to be mapped from VELOOP	To Be Verified
Closed	Account closed	account_status = closed	Stored	Closure vs deletion handling under privacy rules	To Be Verified

6. Data deliberately not collected

Data Not Collected in Analytics	Reason	Alternative
Full name	Not needed to measure behaviour	user_id
Email address / phone number	Direct personal identifiers; not needed for any KPI	registration_method (category only)
Date of birth / exact age	Not needed for the KPIs in scope	Age band only if a real business need is proven
Precise location (GPS, address)	Sensitive and unnecessary	country_code
Raw IP address	Personal data in many regions	Derive country_code, then do not carry the IP into the analytics layer
Advertising ID / device fingerprint	Tracking identifier; not needed unless attribution requires it	campaign_id; revisit with Legal if needed
Government ID or identity documents	Highly sensitive	Not in analytics
Passwords, tokens, API keys, payment credentials	Security secrets; must never appear in analytics	Never collected
Free-text support messages	May contain personal data	issue_category only

7. Open questions for verification

- Does a stable, immutable user_id exist today, and does it survive device changes? (A003)
- Is session tracking present or planned? (A004)
- What are the real account statuses and onboarding steps? (A019, A003)
- Which timestamp format and timezone does the backend use? (A012)
- Which events should qualify as activity, and what activity window should define active vs inactive? (stakeholder decision)

Next: Work Package C - Earning and Reward Analytics.